

(a) Brick Structure Tokenization

Brick Sequence:

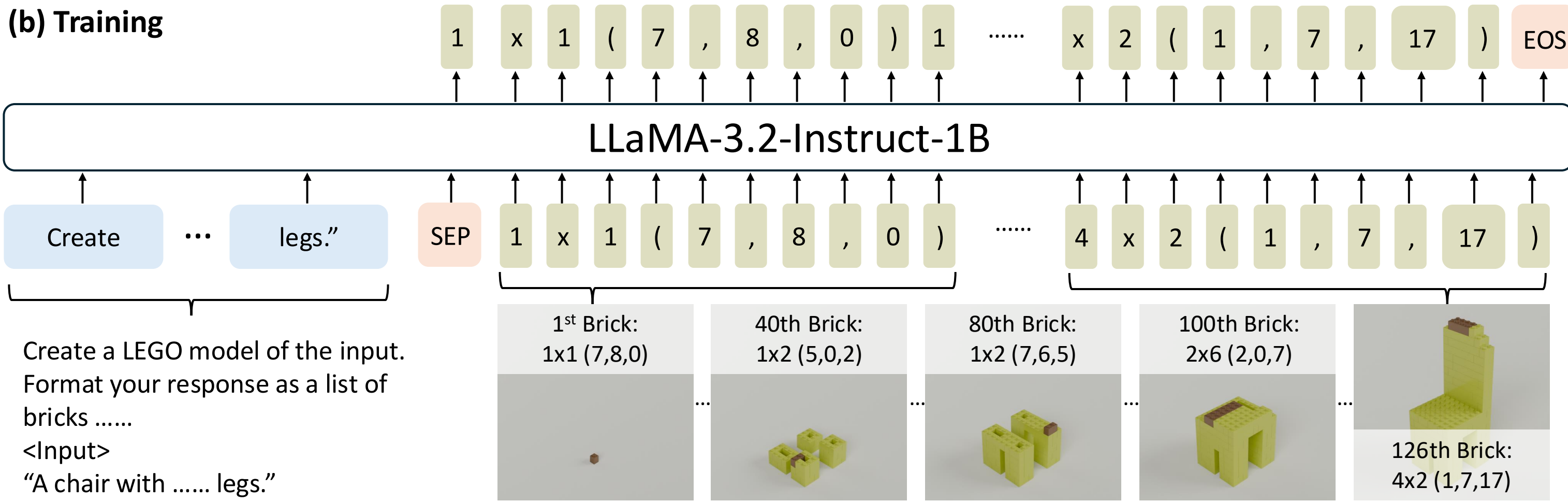
1	1x1 (7,8,0)
2	1x2 (7,6,0)
3	1x1 (7,2,0)
...	...
125	1x2 (5,7,17)
126	4x2 (1,7,17)

Brick Structure:

Each brick *h*x*w* (*x*,*y*,*z*) corresponds to 10 tokens.

Caption:
“A chair with a high, rounded backrest and a flat, rectangular seat supported by four solid, straight legs.”

(b) Training



(c) Inference

User

Create a LEGO model of the input description:
“A rectangular sofa with a high backrest extending to form side armrests, featuring a spacious seating area and a supportive box-like base.”

BrickGPT

Validity Check

- Valid Brick type ✓
- Collision free ✓

Collision ✗

4x6 Brick out of library ✗

Final Stability Check

Output ✓